Extended synthetic wavelength phase imaging by multiwavelength digital holography

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ABSTRACT

We present a technique of visible spectrum quantitative phase imaging with ten millimeter range without phase ambiguity while achieving better than ten micrometer precision. A suitable synthetic wavelength for absolute phase measurement, without unwrapping, is implemented by two-wavelength digital holography. This results in amplified phase noise which is then reduced by an iterative optical and algorithm based process involving one or two additional source wavelengths. We demonstrate that these source wavelengths can be acquired from a single laser diode source by realizing the wavelength output response to fine temperature control.

Keywords: multiwavelength digital holography, quantitative phase, optical phase unwrapping.

1. INTRODUCTION

Digital holographic phase imaging is a precise method of measuring height profiles based on optical path variation; however, it is limited by the wavelength of coherent source illumination[1]. Path differences greater than the wavelength of this illumination will produce wrapped phase profiles. Especially for larger required ranges, this presents a major issue for unwrapping algorithms to efficiently obtain a suitable profile.

We present a technique of visible spectrum quantitative phase profiling with 10-mm range without phase ambiguity. Significantly extended synthetic wavelengths, above millimeter range, can be achieved using two wavelengths with a very small difference (e.g. less than 100 pm); however, the phase noise of the resulting image is magnified by the same factor as the extended range[2,3]. This amplified phase noise is then reduced by an iterative optical and algorithm based process involving, in the present case, one to two additional source wavelengths. These wavelengths are selected to systematically reduce the synthetic wavelength of the previous step. The result of this process has the advantages of both the unambiguous range of the largest synthetic wavelength and the precision of the smallest synthetic wavelength. We demonstrate the application by phase imaging and three-dimensional reconstruction of various appropriately sized objects. Additionally, we demonstrate that our four source wavelengths can be acquired from a single laser diode source by realizing the wavelength output response to fine temperature control.

2. THEORY

A synthetic wavelength height profile is determined by,

$$Z_{ab} = [(\phi_a - \phi_b \bmod 2\pi)/2\pi] \times \Lambda_{ab}$$

where $\varphi_{a,b}$ are the source phase maps and Λ_{ab} is the synthetic wavelength described by,

$$\Lambda_{ab} = \frac{\lambda_a \lambda_b}{\lambda_a - \lambda_b} \ .$$

Then noise reduction is performed by stitching, for example, source wavelength, λ_a, into Z_{ab} as described by the steps,

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$$F = \mathrm{I}\left(\frac{Z_{ab}}{\lambda_a}\right)\lambda_a + z_a$$

$$M = \left|F - Z_{ab}\right| > \frac{\lambda_a}{2}$$

$$M_{+/-} = -\left[\left(F - Z_{ab}\right) > 0\right] \cdot M$$

$$Z_{ab_a} = F + \lambda_a M_{+/-}$$

where the function I() is the integer quotient and M is a logical mask used to determine where it is necessary to add or subtract the value, λ_a. Figure 1 shows an example of this two-wavelength scheme using one of the source phase maps to remove noise from the synthetic map. In this simple 2-micron sloped surface case, noise is easily reduced back to source level in only one step.

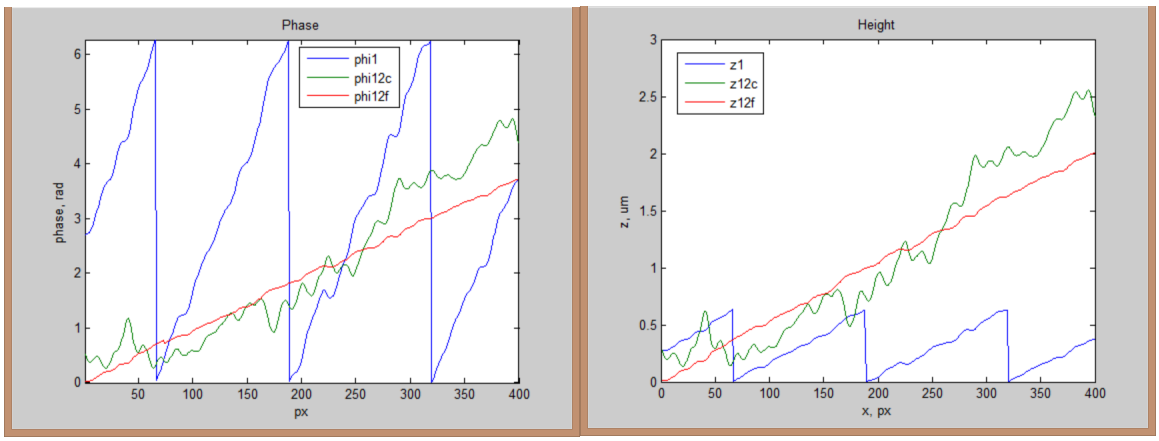

Figure 1. Example cross-section of a smooth sloped object processed using source wavelengths 633nm and 532nm. Blue is source image, green is course unwrapped map, and red is the fine unwrapped map. Phase profiles, left, and height profiles, right.

Increasing the number of source wavelengths and replacing the source wavelength and map in the noise reduction steps with an additional synthetic wavelength and resulting map allows noise to be incrementally removed. Each additional synthetic wavelength must be larger than the noise level of the previous step in order to be effective. In our experiments, we achieve desired wavelengths using a single laser diode source which is tuned by temperature control (figure 2). Typically, temperature control was able to yield results consistent enough for our purposes. We were able to select wavelengths with a difference as small as 20 pm, producing a synthetic wavelength of over 20 mm, the requirement for our goal of 10-mm measurement in reflection mode.

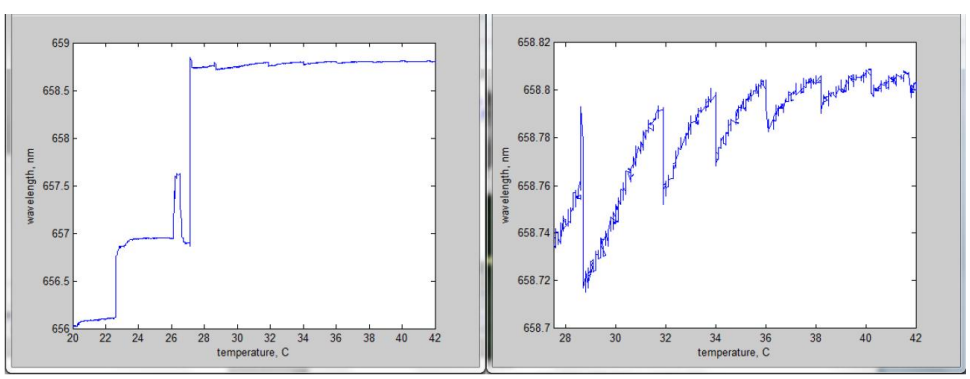

Figure 2. Temperature dependence of wavelength for a single longitudinal mode laser diode. A laser module is temperature controlled to achieve desired wavelengths.

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3. EXPERIMENTS

3.1 Specular objects

Figure 3 shows an example of the technique applied to a chromium deposited sample siting a top a larger flat mirror. The height of this sample (chromium surface from mirror) was previously measured physically by a caliper micrometer at approximately 600 um as is reasonably suggested by our phase measurement as well.

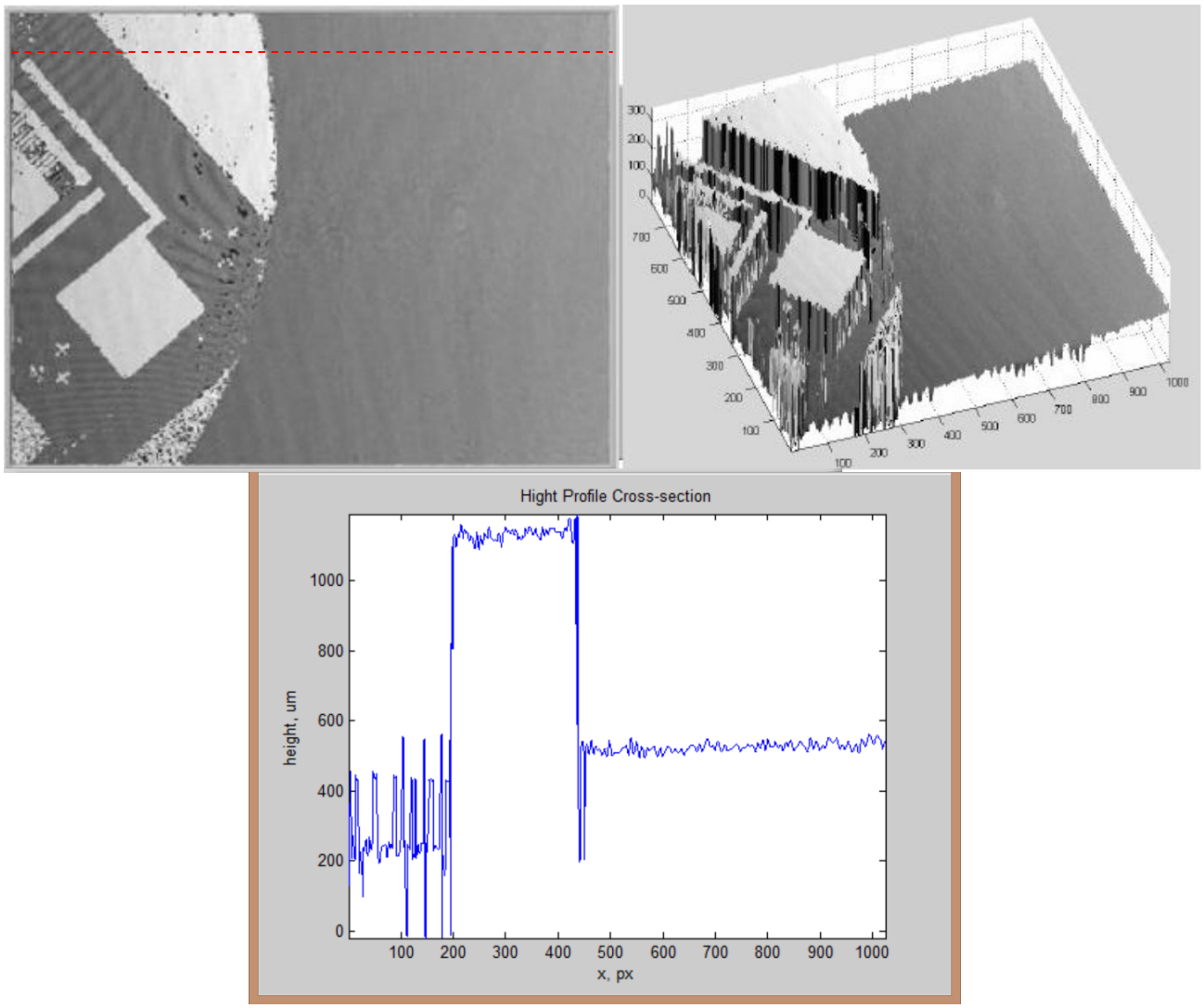

Figure 3. Height profile of chromium deposited sample on flat mirror background with 3D rendering and cross-section indicated by red dashed line. Height of chromium deposit expected ~600um.

We next sought to test an object with an expected height closer to our goal range of 10 mm. Figure 4 demonstrates our phase measurement capabilities on a round flat mirror of height 5.9 mm attached to a larger flat mirror base.

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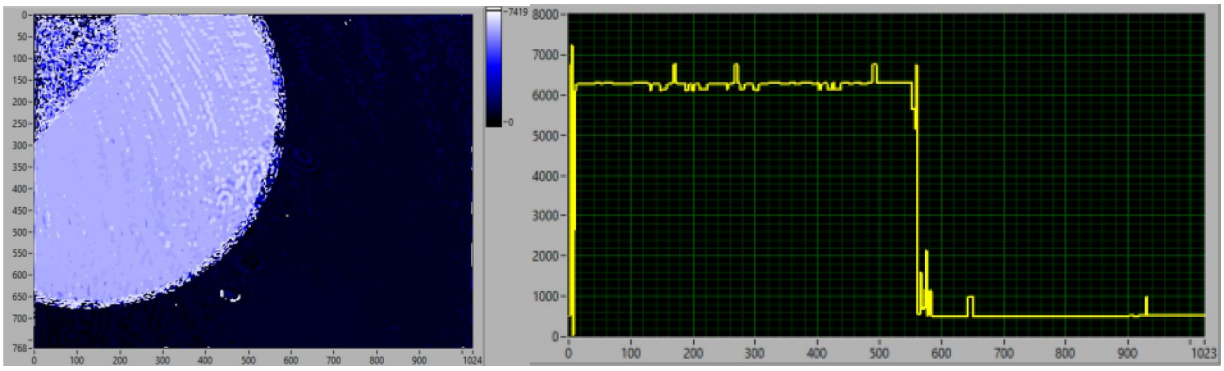

Figure 4. Extended height object. Round flat mirror of 5.9 mm height on larger flat mirror.

3.2 Diffuse objects

We have also attempted application of our process to diffuse objects which required some additional parameters. Firstly, our imaging system required a special filter (an iris) applied at the Fourier plane in order to preserve the angular spectrum for adequate numerical filtering. The main issue addressed by this is the parasitic interference from source points outside of the intended field of view. Although speckle noise (and other possible culprits) still made complete noise reduction by multiwavelength optical phase unwrapping difficult to achieve, we were able to improve results by applying a software median filter as demonstrated on the face of a U.S. Quarter in figures 5 and 6.

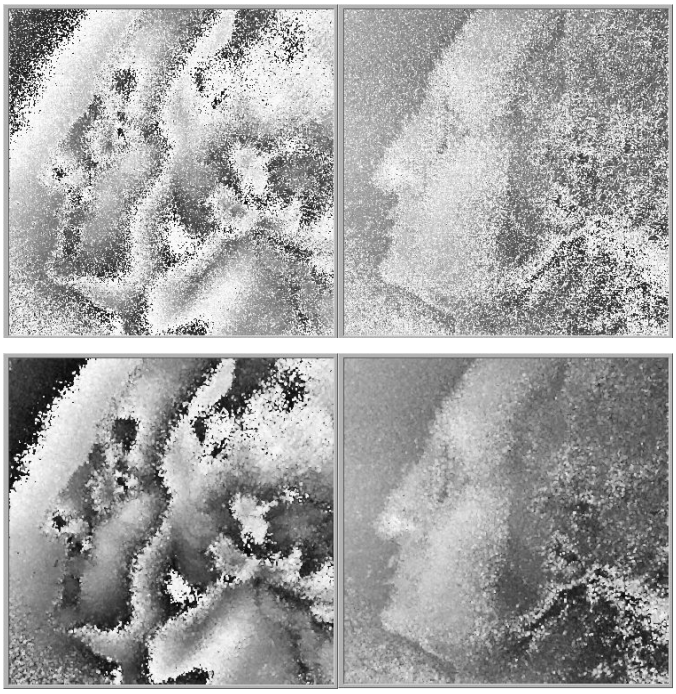

Figure 5. Multiwavelength unwrapping of diffuse surface of U.S. Quarter. Top: a wrapped intermediate step and final map of standard process described above. Bottom: result of applying a software median filter of rank 3 to the intermediate steps.

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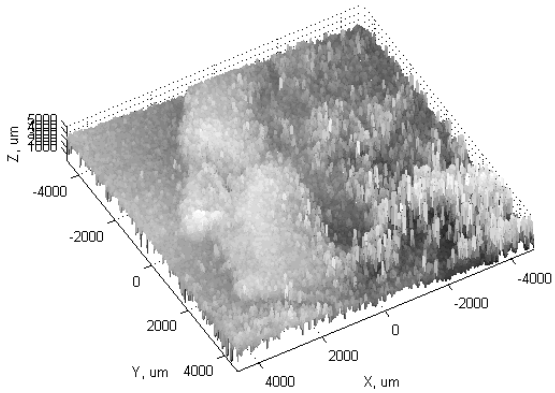

Figure 6. 3D rendering of the coin from figure 5.

4. CONCLUSION

We were successfully able to synthesize wavelengths in the 20-mm range allowing us to phase profile objects without ambiguities within the range of 10 mm in reflection mode. We have demonstrated significant reduction of the resulting amplified noise using additional source wavelengths. Additionally, we have demonstrated some success in profiling diffusive surfaces with the addition of Fourier plane spatial filtering and software median filtering.

REFERENCES

[1] Kim, M.K., "Principles and techniques of digital holographic microscopy," SPIE Reviews 1, 018005-1-50 (2010).
[2] Wagner, C., Osten, W., and Seebacher, S., "Direct shape measurement by digital wavefront reconstruction and multiwavelength contouring," Opt. Eng. 39(1), 79-85 (2000).
[3] Warnasooriya, N. and Kim, M.K., "Quantitative phase imaging using three-wavelength optical phase unwrapping," J. Mod. Opt. 56(1), 67-74 (2010).

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